Android Application Development Lab

Beginner level Experiments

Name : Aman Kumar Akhil

Roll No. - 25/SCA/MCAN/059

Q1 -> Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

Ans1 ->

Xml –

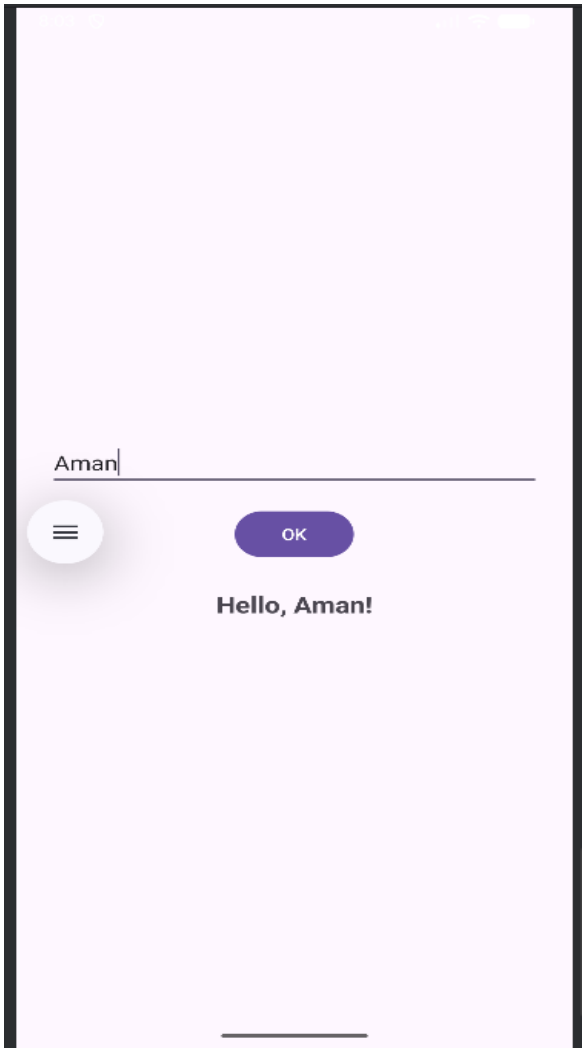
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center">

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:inputType="textPersonName"/>

    <Button
        android:id="@+id/btnOK"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="OK"/>

    <TextView
        android:id="@+id/tvHello"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:textSize="20sp"
        android:textStyle="bold"
        android:text=""/>

</LinearLayout>
```

Java:

```java
package com.example.okbutton;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    2 usages
    EditText etName;
    2 usages
    Button btnOK;
    3 usages
    TextView tvHello;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName  = findViewById(R.id.etName);
        btnOK   = findViewById(R.id.btnOK);
        tvHello = findViewById(R.id.tvHello);

        btnOK.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = etName.getText().toString().trim();
                if (!name.isEmpty()) {
```

```java
                if (!name.isEmpty()) {
                    tvHello.setText("Hello, " + name + "!");
                } else {
                    tvHello.setText("Please enter a name.");
                }
            }
        });
    }
}
```

Output:



Q2 -> Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button. Use 1. Linear Layout, 2. Relative Lavout and 3. Grid Layout or Table Layout.

Ans2 ->

Xml:

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:padding="14dp">

        <TextView android:text="User Name" android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
        <EditText android:id="@+id/etName" android:layout_width="match_parent"
            android:layout_height="wrap_content" android:hint="Enter name"/>

        <TextView android:text="Password" android:layout_width="wrap_content"
            android:layout_height="wrap_content" android:layout_marginTop="6dp"/>
        <EditText android:id="@+id/etPassword"
            android:layout_width="match_parent" android:layout_height="wrap_content"
            android:inputType="textPassword" android:hint="Enter password"/>

        <TextView android:text="Address" android:layout_width="wrap_content"
            android:layout_height="wrap_content" android:layout_marginTop="6dp"/>
        <EditText android:id="@+id/etAddress"
            android:layout_width="match_parent" android:layout_height="wrap_content"
            android:hint="Enter address"/>

        <TextView android:text="Gender" android:layout_width="wrap_content"
            android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
        <RadioGroup android:id="@+id/rgGender" android:orientation="horizontal"
            android:layout_width="wrap_content" android:layout_height="wrap_content">
```

```xml
            android:layout_width="wrap_content" android:layout_height="wrap_content"/>
        <RadioButton android:id="@+id/rbFemale" android:text="Female"
            android:layout_width="wrap_content" android:layout_height="wrap_content"/>
    </RadioGroup>

    <TextView android:text="Age" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:layout_marginTop="6dp"/>
    <EditText android:id="@+id/etAge" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:inputType="number"
        android:hint="Enter age"/>

    <TextView android:text="Date of Birth"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_marginTop="6dp"/>
    <DatePicker android:id="@+id/dpDOB" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:calendarViewShown="false"/>

    <TextView android:text="State" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:layout_marginTop="4dp"/>
    <Spinner android:id="@+id/spState" android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button android:id="@+id/btnSubmit" android:layout_width="match_parent"
        android:layout_height="wrap_content" android:layout_marginTop="2dp"
        android:text="Submit"/>

    <TextView android:id="@+id/tvResult"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_marginTop="16dp" android:textSize="15sp"/>
```

```xml
    </LinearLayout>
</ScrollView>
```

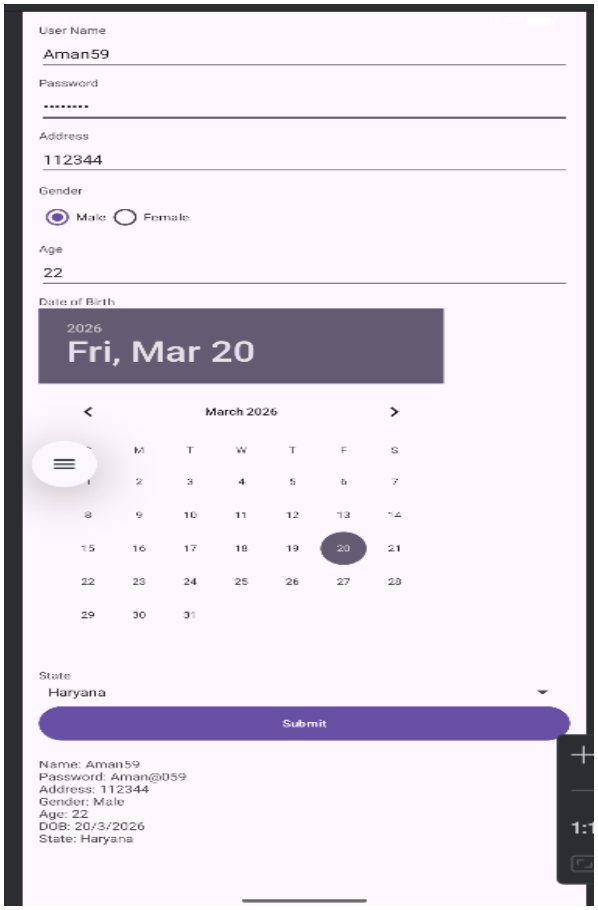
Java:

```java
package com.example.formapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    2 usages
    EditText etName, etPassword, etAddress, etAge;
    2 usages
    RadioGroup rgGender;
    4 usages
    DatePicker dpDOB;
    3 usages
    Spinner spState;
    2 usages
    Button btnSubmit;
    2 usages
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); // swap XML here

        etName      = findViewById(R.id.etName);
        etPassword  = findViewById(R.id.etPassword);
        etAddress   = findViewById(R.id.etAddress);
        etAge       = findViewById(R.id.etAge);
        rgGender    = findViewById(R.id.rgGender);
        dpDOB       = findViewById(R.id.dpDOB);
```

```java
        spState     = findViewById(R.id.spState);
        btnSubmit   = findViewById(R.id.btnSubmit);
        tvResult    = findViewById(R.id.tvResult);

        // Populate spinner
        String[] states = {"Delhi", "Haryana", "UP", "Maharashtra",
                "Karnataka"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>( context: this,
                android.R.layout.simple_spinner_item, states);

        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item)
        ;
        spState.setAdapter(adapter);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name     = etName.getText().toString();
                String password = etPassword.getText().toString();
                String address  = etAddress.getText().toString();
                String age      = etAge.getText().toString();
                String state    = spState.getSelectedItem().toString();

                int selectedGenderId = rgGender.getCheckedRadioButtonId();
                RadioButton rbGender = findViewById(selectedGenderId);
                String gender = (rbGender != null) ?
                        rbGender.getText().toString() : "Not selected";

                String dob = dpDOB.getDayOfMonth() + "/" +
                        (dpDOB.getMonth() + 1) + "/" +
```

```java
                int selectedGenderId = rgGender.getCheckedRadioButtonId();
                RadioButton rbGender = findViewById(selectedGenderId);
                String gender = (rbGender != null) ?
                        rbGender.getText().toString() : "Not selected";

                String dob = dpDOB.getDayOfMonth() + "/" +
                        (dpDOB.getMonth() + 1) + "/" +
                        dpDOB.getYear();

                tvResult.setText(
                        "Name: "      + name     + "\n" +
                        "Password: "  + password + "\n" +
                        "Address: "   + address  + "\n" +
                        "Gender: "    + gender   + "\n" +
                        "Age: "       + age      + "\n" +
                        "DOB: "       + dob      + "\n" +
                        "State: "     + state
                );
            }
        });
    }
}
```

Output:



Q3 -> Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

Ans3 ->

Xml –

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp">

    <EditText
        android:id="@+id/etTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Notification Title" />

    <EditText
        android:id="@+id/etMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:hint="Notification Message" />

    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Send Notification" />

</LinearLayout>
```

Java-

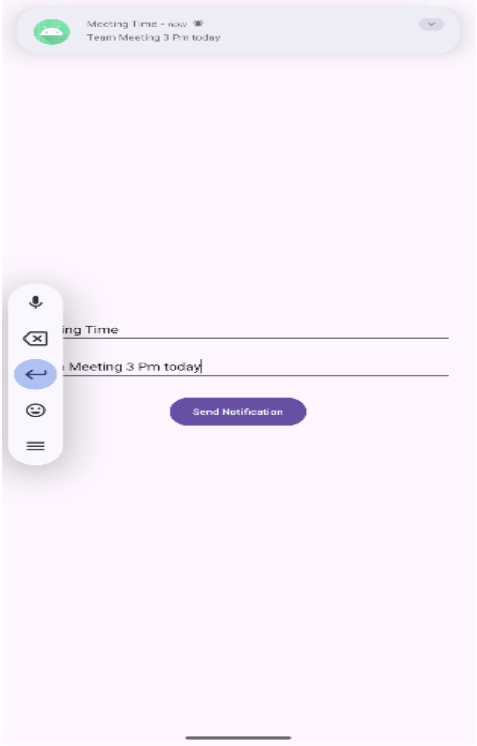
```java
package com.example.notificationsent;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import androidx.core.content.ContextCompat;

import android.Manifest;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    2 usages
    EditText etTitle, etMessage;
    2 usages
    Button btnSend;

    2 usages
    private static final String CHANNEL_ID = "my_channel";
    1 usage
    private static int notificationId = 0;
```

```java
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    etTitle   = findViewById(R.id.etTitle);
    etMessage = findViewById(R.id.etMessage);
    btnSend   = findViewById(R.id.btnSend);

    createNotificationChannel();

    btnSend.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            String title   = etTitle.getText().toString().trim();
            String message = etMessage.getText().toString().trim();

            if (title.isEmpty() || message.isEmpty()) {
                Toast.makeText( context: MainActivity.this,
                        text: "Please enter title and message",
                        Toast.LENGTH_SHORT).show();
                return;
            }

            // Check for POST_NOTIFICATIONS permission on Android 13+
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
```

```java
if (ContextCompat.checkSelfPermission( context: MainActivity.this, Manifest.permission.POST_NOTIFICATIONS) != PackageManager.PERMISSION_GRANTED) {
    requestPermissions(new String[]{Manifest.permission.POST_NOTIFICATIONS}, requestCode: 101);
    return;
}



Build and show notification
tificationCompat.Builder builder = new
    NotificationCompat.Builder(
    context: MainActivity.this, CHANNEL_ID)
    .setSmallIcon(android.R.drawable.ic_dialog_info)
    .setContentTitle(title)
    .setContentText(message)
    .setPriority(NotificationCompat.PRIORITY_HIGH)
    .setAutoCancel(true);

tificationManagerCompat notificationManager =
    NotificationManagerCompat.from( context: MainActivity.this);
y {
  notificationManager.notify(notificationId++, builder.build());
catch (SecurityException e) {
  Toast.makeText( context: MainActivity.this, text: "Permission denied for notifications", Toast.LENGTH_SHORT).show();
```

```
                // Show Toast with notification details
                Toast.makeText( context: MainActivity.this,
                        text: "Notification Sent!\nTitle: " + title + "\nMessage: " +
                                message,
                        Toast.LENGTH_LONG).show();
            }
        });
    }

    1 usage
    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            CharSequence name = "My Channel";
            String description = "Channel for app notifications";
            int importance = NotificationManager.IMPORTANCE_HIGH;
            NotificationChannel channel = new NotificationChannel(CHANNEL_ID,
                    name, importance);
            channel.setDescription(description);
            NotificationManager notificationManager =
                    getSystemService(NotificationManager.class);
            if (notificationManager != null) {
                notificationManager.createNotificationChannel(channel);
            }
        }
    }
```

Output-



Q4 -> Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.

Ans4 ->

Xml –

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        android:background="?attr/colorPrimary"
        android:title="Q4 Menu App"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:text="Press the 3-dot menu (top right)"
        android:textSize="18sp"/>

</LinearLayout>
```

Java-

```java
package com.example.intents;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
    }

    // Inflate the menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }

    // Handle menu item clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();
```

```java
        if (id == R.id.menu_dial) {
            // Dial a number
            Intent dialIntent = new Intent(Intent.ACTION_DIAL);
            dialIntent.setData(Uri.parse( uriString: "tel:9876543210"));
            startActivity(dialIntent);
            return true;

        } else if (id == R.id.menu_website) {
            // Open a website
            Intent webIntent = new Intent(Intent.ACTION_VIEW);
            webIntent.setData(Uri.parse( uriString: "https://www.google.com"));
            startActivity(webIntent);
            return true;

        } else if (id == R.id.menu_sms) {
            // Send an SMS
            Intent smsIntent = new Intent(Intent.ACTION_SENDTO);
            smsIntent.setData(Uri.parse( uriString: "smsto:9876543210"));
            smsIntent.putExtra( name: "sms_body",  value: "Hello! This is a test SMS.");
            startActivity(smsIntent);
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

Output-

Q5 -> Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message.

Ans5 ->

Xml -

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"/>

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username"
        android:inputType="textPersonName"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="12dp"/>
```

```
    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Login"/>

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:text="Register (Save User)"/>

</LinearLayout>
```

Java-

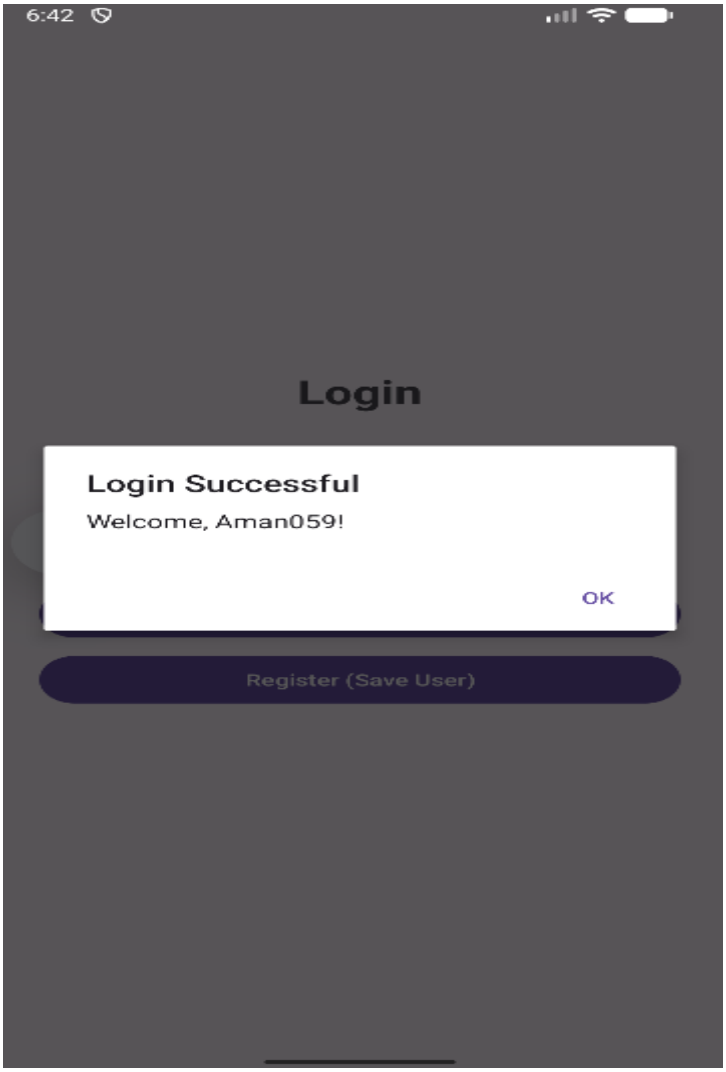
```
package com.example.loginsuccessfulandfailed;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import java.io.*;

public class MainActivity extends AppCompatActivity {

    3 usages
    EditText etUsername, etPassword;
    2 usages
    Button btnLogin, btnRegister;
    2 usages
    private static final String FILE_NAME = "users.txt";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUsername  = findViewById(R.id.etUsername);
        etPassword  = findViewById(R.id.etPassword);
        btnLogin    = findViewById(R.id.btnLogin);
        btnRegister = findViewById(R.id.btnRegister);
```

```java
// Save new user to text file
btnRegister.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String username = etUsername.getText().toString().trim();
        String password = etPassword.getText().toString().trim();

        if (username.isEmpty() || password.isEmpty()) {
            Toast.makeText( context: MainActivity.this,
                    text: "Please enter username and password",
                    Toast.LENGTH_SHORT).show();
            return;
        }

        try {
            FileOutputStream fos = openFileOutput(FILE_NAME,
                    MODE_APPEND);
            String record = username + "\t" + password + "\n";
            fos.write(record.getBytes());
            fos.close();
            Toast.makeText( context: MainActivity.this,
                    text: "User registered successfully!",
                    Toast.LENGTH_SHORT).show();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
});

// Verify login from text file
btnLogin.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String username = etUsername.getText().toString().trim();
        String password = etPassword.getText().toString().trim();

        if (username.isEmpty() || password.isEmpty()) {
            Toast.makeText( context: MainActivity.this,
                    text: "Please enter username and password",
                    Toast.LENGTH_SHORT).show();
            return;
        }

        boolean loginSuccess = false;

        try {
            FileInputStream fis = openFileInput(FILE_NAME);
            BufferedReader reader = new BufferedReader(new
                    InputStreamReader(fis));
            String line;

            while ((line = reader.readLine()) != null) {
                String[] parts = line.split( regex: "\t");
                if (parts.length == 2) {
                    String savedUser = parts[0].trim();
                    String savedPass = parts[1].trim();
```

```
                    if (savedUser.equals(username) &&
                            savedPass.equals(password)) {
                        loginSuccess = true;
                        break;
                    }
                }
            }
            reader.close();
            fis.close();

        } catch (FileNotFoundException e) {
            showDialog( title: "Login Failed",  message: "No users registered yet!");
            return;
        } catch (IOException e) {
            e.printStackTrace();
        }

        if (loginSuccess) {
            showDialog( title: "Login Successful",  message: "Welcome, " + username +
                    "!");
        } else {
            showDialog( title: "Login Failed",  message: "Invalid username or password.");
        }
    }
});
}
```

```
    private void showDialog(String title, String message) {
        new AlertDialog.Builder( context: this)
                .setTitle(title)
                .setMessage(message)
                .setPositiveButton( text: "OK",  listener: null)
                .show();
    }
}
```

Output-



Q6 -> Create an activity to navigate to different activities through Intents- Implicit and Explicit.

Ans6 ->

Xml –

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Intent Navigation Demo"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="32dp"/>
```

```xml
    <Button
        android:id="@+id/btnExplicit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Explicit Intent (Go to Second Activity)"/>

    <Button
        android:id="@+id/btnImplicitWeb"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Implicit Intent (Open Website)"/>

    <Button
        android:id="@+id/btnImplicitCall"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Implicit Intent (Dial Number)"/>

    <Button
        android:id="@+id/btnImplicitMap"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Implicit Intent (Open Map)"/>

</LinearLayout>
```

Xml (acitivity_second.xml) –

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity!"
        android:textSize="22sp"
        android:textStyle="bold"/>

    <Button
        android:id="@+id/btnBack"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:text="Go Back"/>

</LinearLayout>
```

Java (MainActivity.java) –

```java
package com.example.intents_implicitandexplicit;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    2 usages
    Button btnExplicit, btnImplicitWeb, btnImplicitCall, btnImplicitMap;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnExplicit     = findViewById(R.id.btnExplicit);
        btnImplicitWeb  = findViewById(R.id.btnImplicitWeb);
        btnImplicitCall= findViewById(R.id.btnImplicitCall);
        btnImplicitMap  = findViewById(R.id.btnImplicitMap);

        // EXPLICIT INTENT — go to MainActivity2
        btnExplicit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent( packageContext: MainActivity.this,
                        MainActivity2.class);
                intent.putExtra( name: "message",  value: "Hello from MainActivity!");
                startActivity(intent);
```

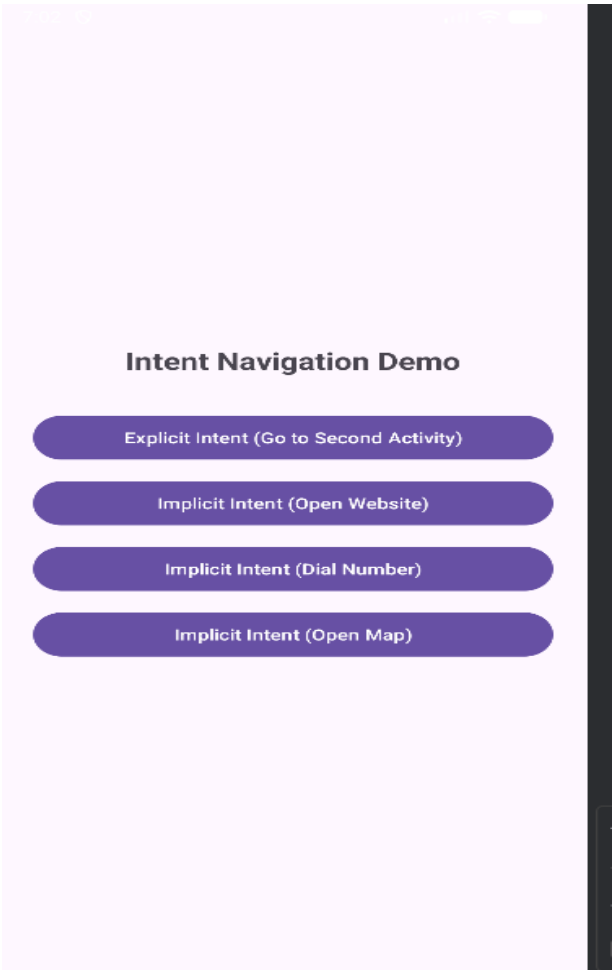
```java
            startActivity(intent);
        }
    });

    // IMPLICIT INTENT — open website
    btnImplicitWeb.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse( uriString: "https://www.google.com"));
            startActivity(intent);
        }
    });

    // IMPLICIT INTENT — dial number
    btnImplicitCall.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse( uriString: "tel:9876543210"));
            startActivity(intent);
        }
    });

    // IMPLICIT INTENT — open map
    btnImplicitMap.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
```

```java
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse( uriString: "geo:28.6139,77.2090?q=New+Delhi"));
            startActivity(intent);
        }
    });
}
}
```

Java (SecondActivity.java) –

```java
package com.example.intents_implicitandexplicit;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);

        TextView tvMessage = findViewById(R.id.tvMessage);
        Button btnBack     = findViewById(R.id.btnBack);

        // Receive data from MainActivity
        String message = getIntent().getStringExtra( name: "message");
        if (message != null) {
            tvMessage.setText(message);
        }

        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                finish(); // goes back to MainActivity
            }
        });
    }
}
```

Output-



Q7 -> Create an activity to create option Menu.

Ans7 ->

Xml (activity_main.xml) –

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select an option from the menu"
        android:textSize="20sp"
        android:gravity="center"
        android:textStyle="bold"/>
</LinearLayout>
```

Xml (menu_options.xml) –

```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menu_home"
        android:title="Home"
        android:icon="@android:drawable/ic_menu_today"
        android:showAsAction="ifRoom"/>

    <item
        android:id="@+id/menu_search"
        android:title="Search"
        android:icon="@android:drawable/ic_menu_search"
        android:showAsAction="ifRoom"/>

    <item
        android:id="@+id/menu_settings"
        android:title="Settings"
        android:icon="@android:drawable/ic_menu_preferences"
        android:showAsAction="never"/>

    <item
        android:id="@+id/menu_share"
        android:title="Share"
        android:icon="@android:drawable/ic_menu_share"
        android:showAsAction="never"/>

    <item
        android:id="@+id/menu_about"
        android:title="About"
        android:icon="@android:drawable/ic_menu_info_details"
        android:showAsAction="never"/>
```

```xml
    <item
        android:id="@+id/menu_exit"
        android:title="Exit"
        android:icon="@android:drawable/ic_menu_close_clear_cancel"
        android:showAsAction="never"/>

</menu>
```

Main.Activity.java

```java
package com.example.optmenu;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    6 usages
    private TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable( $this$enableEdgeToEdge: this);
        setContentView(R.layout.activity_main);

        tvResult = findViewById(R.id.tvResult);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), ( View v, WindowInsetsCompat insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
```

```java
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu resource. Since your menu code was in activity_main2.xml,
        // it's better to move it to res/menu/options_menu.xml.
        // I'll assume it's available as R.menu.options_menu.
        getMenuInflater().inflate(R.menu.options_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.menu_home) {
            tvResult.setText("Home Selected");
            return true;
        } else if (id == R.id.menu_search) {
            tvResult.setText("Search Selected");
            return true;
        } else if (id == R.id.menu_settings) {
```

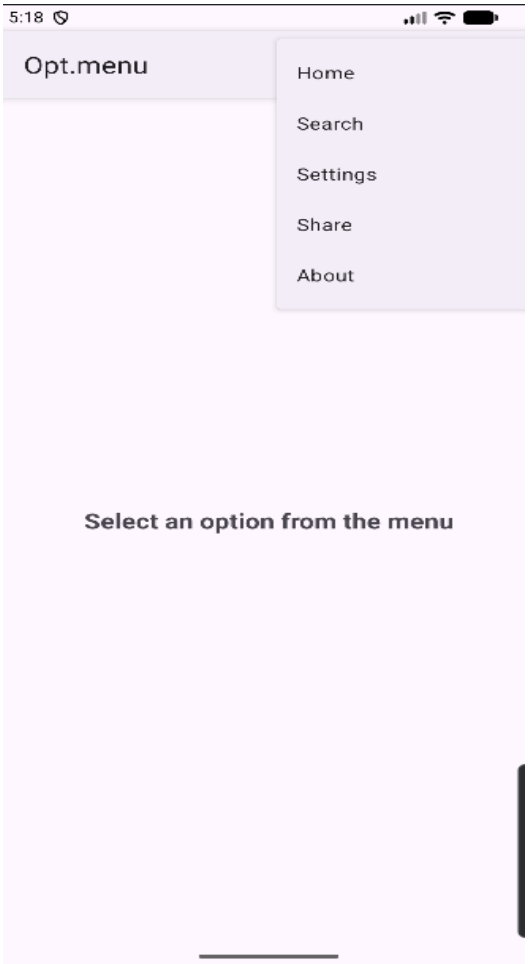
```
                tvResult.setText("Settings Selected");
                return true;
            } else if (id == R.id.menu_share) {
                tvResult.setText("Share Selected");
                return true;
            } else if (id == R.id.menu_about) {
                tvResult.setText("About Selected");
                return true;
            }

            return super.onOptionsItemSelected(item);
        }
    }
```

MainActivity2.java

```
package com.example.optmenu;

import ...

public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.options_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.menu_home) {
            Toast.makeText( context: this,  text: "Home selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_search) {
            Toast.makeText( context: this,  text: "Search selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_settings) {
            Toast.makeText( context: this,  text: "Settings selected", Toast.LENGTH_SHORT).show();
```

```
            return true;
        } else if (id == R.id.menu_share) {
            Toast.makeText( context: this,  text: "Share selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_about) {
            Toast.makeText( context: this,  text: "About selected", Toast.LENGTH_SHORT).show();
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

Output:



Q8-> Create an activity using Button Controls- CheckBox, RadioButtons, Radio Groups.

Xml:

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Registration Form"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent" />

    <CheckBox
        android:id="@+id/checkBox"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:text="I agree to the Terms and Conditions"
        app:layout_constraintTop_toBottomOf="@id/title"
        app:layout_constraintStart_toStartOf="parent" />
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:text="Select your gender:"
        android:textSize="16sp"
        app:layout_constraintTop_toBottomOf="@id/checkBox"
        app:layout_constraintStart_toStartOf="parent" />

    <RadioGroup
        android:id="@+id/radioGroup"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        app:layout_constraintTop_toBottomOf="@id/radioLabel"
        app:layout_constraintStart_toStartOf="parent">

        <RadioButton
            android:id="@+id/radioMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male" />

        <RadioButton
            android:id="@+id/radioFemale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female" />
```

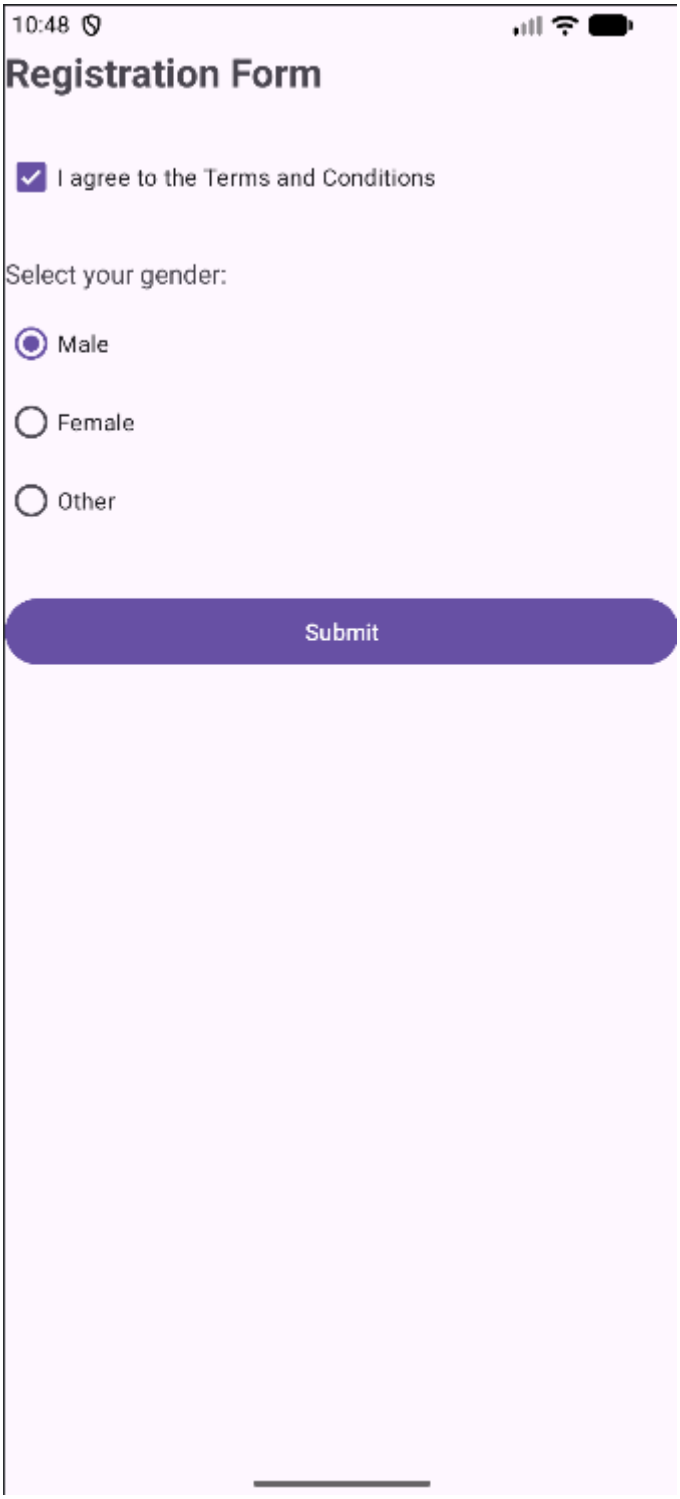
```
        <RadioButton
            android:id="@+id/radioOther"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Other" />

    </RadioGroup>

    <Button
        android:id="@+id/submitButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="Submit"
        app:layout_constraintTop_toBottomOf="@id/radioGroup"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.Java:

```java
package com.example.myapplication;

import ...

public class MainActivity extends AppCompatActivity {

    2 usages
    private CheckBox checkBox;
    2 usages
    private RadioGroup radioGroup;
    2 usages
    private Button submitButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable( $this$enableEdgeToEdge: this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), ( View v,  WindowInsetsCompat insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        // Initialize UI components
        checkBox = findViewById(R.id.checkBox);
        radioGroup = findViewById(R.id.radioGroup);
        submitButton = findViewById(R.id.submitButton);
```

```java
        submitButton.setOnClickListener( View v -> {
            boolean isChecked = checkBox.isChecked();
            int selectedId = radioGroup.getCheckedRadioButtonId();

            String gender = "None";
            if (selectedId != -1) {
                RadioButton selectedRadioButton = findViewById(selectedId);
                gender = selectedRadioButton.getText().toString();
            }

            String message = "Agreed: " + isChecked + "\nGender: " + gender;
            Toast.makeText( context: MainActivity.this, message, Toast.LENGTH_SHORT).show();
        });
    }
}
```

Output:


10:48
Registration Form
I agree to the Terms and Conditions
Select your gender:
Male
Female
Other
Submit